Complete Lattices. Fixed points

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Lattices

Definition: A lattice is a partial order in which every two-element set has a least upper bound and a greatest lower bound (so, we have $\sqcap$ and $\sqcup$ as well-defined binary operations).

Note: this means every non-empty *finite* set of elements $\{x_1, \ldots, x_n\}$ has:

- least upper bound $x_1 \sqcup \ldots \sqcup x_n$
- greatest lower bound $x_1 \sqcap \ldots \sqcap x_n$

In general infinite sets in a lattice need not have lub or glb.

Complete Semilattice is a Complete Lattice

If we have all $\sqcap$-s we then also have all $\sqcup$-s:
Theorem: Let $(A, \sqsubseteq)$ be a partial order such that every set $S \subseteq A$ (including infinite sets) has the greatest lower bound ($\sqcap$). Prove that then every set $S \subseteq A$ has the least upper bound ($\sqcup$).

Example: Application of the Previous Theorem

Let U be a set and $A \subseteq U \times U$ the set of all **equivalence relations** on this set. Consider the partial order $(A, \subseteq)$.

Lemma

If $I \subseteq A$ is a set of equivalence relations, then $\cap I$ is also an equivalence relation.

Consequence: Given $I \subseteq A$ there exists the least equivalence relation containing every relation from I (equivalence closure of relations in I).

Note: **congruence** is equivalence relation that agrees with some operations. For example, $x \sim x'$ and $y \sim y'$ implies $(x + y) \sim (x' + y')$. The analogous properties hold for congruence relations.

Complete Lattices

Definition: A **complete** lattice is a lattice where for every set S (including empty set and infinite sets) there exist $\sqcup S$ and $\sqcap S$.

Note that:

- a complete lattice has the least element of the lattice (usually denoted $\perp$) and the greatest element of the lattice (usually denoted $\top$)
- every finite lattice is complete

Fixpoints

Definition: Given a set A and a function $f : A \to A$ we say that $x \in A$ is a fixed point (fixpoint) of f if $f(x) = x$.

Definition: Let $(A, \leq)$ be a partial order, let $f : A \to A$ be a monotonic function on $(A, \leq)$, and let the set of its fixpoints be $S = \{x \mid f(x) = x\}$. If the least element of S exists, it is called the **least fixpoint**, if the greatest element of S exists, it is called the **greatest fixpoint**.

Fixpoints

Let $(A, \sqsubseteq)$ be a complete lattice and $G : A \to A$ a monotonic function.

Definition:
$\mathrm{Post} = \{x \mid G(x) \sqsubseteq x\}$ - the set of *postfix points* of G
(e.g. $\top$ is a postfix point)
$\mathrm{Pre} = \{x \mid x \sqsubseteq G(x)\}$ - the set of *prefix points* of G
$\mathrm{Fix} = \{x \mid G(x) = x\}$ - the set of *fixed points* of G.

Note that $\mathrm{Fix} \subseteq \mathrm{Post}$.
Note also:

- $\perp \in \mathrm{Pre}$. Also, $x \in \mathrm{Pre}$ implies $G(x) \in \mathrm{Pre}$. Hence $G^i(\perp) \in \mathrm{Pre}$ for all $i \in \mathbb{N}$
- $\top \in \mathrm{Post}$. Also, $x \in \mathrm{Post}$ implies $G(x) \in \mathrm{Post}$. Hence $G^i(\top) \in \mathrm{Post}$ for all $i \in \mathbb{N}$

Tarski's fixed point theorem

Theorem: Let $a = \sqcap \mathrm{Post}$. Then a is the least element of Fix (dually, $\sqcup \mathrm{Pre}$ is the largest element of Fix).

Proof:

Let x range over elements of Post.

- applying monotonic G from $a \sqsubseteq x$ we get $G(a) \sqsubseteq G(x) \sqsubseteq x$
- so $G(a)$ is a lower bound on Post, but a is the greatest lower bound, so $G(a) \sqsubseteq a$
- therefore $a \in \mathrm{Post}$
- Post is closed under G, by monotonicity, so $G(a) \in \mathrm{Post}$
- a is a lower bound on Post, so $a \sqsubseteq G(a)$
- from $a \sqsubseteq G(a)$ and $G(a) \sqsubseteq a$ we have $a = G(a)$, so $a \in \mathrm{Fix}$
- a is a lower bound on Post so it is also a lower bound on a smaller set Fix

In fact, the set of all fixpoints Fix is a lattice itself.

Tarski's fixed point theorem

Tarski's Fixed Point theorem shows that in a complete lattice with a monotonic function G on this lattice, there is at least one fixed point of G, namely the least fixed point $\sqcap$Post.

- Tarski's theorem guarantees fixpoints in complete lattices, but the above proof does not say how to find them.
- How difficult it is to find fixpoints depends on the structure of the lattice.

Let G be a monotonic function on a lattice with a bottom element. Let $a_0 = \bot$ and $a_{n+1} = G(a_n)$. We obtain a sequence $\bot \sqsubseteq G(\bot) \sqsubseteq G^2(\bot) \sqsubseteq \cdots$. Let $a_* = \bigsqcup_{n \geq 0} a_n$.

Lemma: The value a_* is a prefix point. Proof hint: $G(G^n(\bot)) \sqsubseteq G(a_*)$

Observation: a_* need not be a fixpoint. Example: take $f : [0,1] \to [0,1]$,

$$f(x) = \begin{cases} \frac{x+1}{2}, & \text{if } 0 \leq x < 1 \\ 2, & \text{if } x \geq 1 \end{cases}$$

Omega continuity

Definition: A function G is ω-continuous if for every chain $x_0 \sqsubseteq x_1 \sqsubseteq \ldots \sqsubseteq x_n \sqsubseteq \ldots$ such that $\bigsqcup_{i \geq 0} x_i$ exists, we have that $\bigsqcup_{i \geq 0} G(x_i)$ exists and

$$G(\bigsqcup_{i \geq 0} x_i) = \bigsqcup_{i \geq 0} G(x_i)$$

Lemma: For an ω-continuous function G, the value $a_* = \bigsqcup_{n \geq 0} G^n(\bot)$ is the least fixpoint of G.
(Proof on the next slide.)

Iterating sequences and omega continuity

Lemma: For an ω-continuous function G, the value $a_* = \bigsqcup_{n \geq 0} G^n(\perp)$ is the least fixpoint of G. (Special case of (W) Kleene fixed-point theorem.)

Proof:

- By definition of ω-continuous we have $G(\bigsqcup_{n \geq 0} G^n(\perp)) = \bigsqcup_{n \geq 0} G^{n+1}(\perp) = \bigsqcup_{n \geq 1} G^n(\perp)$.
- But $\bigsqcup_{n \geq 0} G^n(\perp) = \bigsqcup_{n \geq 1} G^n(\perp) \sqcup \perp = \bigsqcup_{n \geq 1} G^n(\perp)$ because $\perp$ is the least element of the lattice.
- Thus $G(\bigsqcup_{n \geq 0} G^n(\perp)) = \bigsqcup_{n \geq 0} G^n(\perp)$ and a_* is a fixpoint.

Now let's prove it is the least. Let c be such that $G(c) = c$. We want $\bigsqcup_{n \geq 0} G^n(\perp) \sqsubseteq c$. This is equivalent to $\forall n \in \mathbb{N}. G^n(\perp) \sqsubseteq c$.
We can prove this by induction : $\perp \sqsubseteq c$ and if $G^n(\perp) \sqsubseteq c$, then by monotonicity of G and by definition of c we have $G^{n+1}(\perp) \sqsubseteq G(c) \sqsubseteq c$.

Iterating sequences and omega continuity

So, for an ω-continuous function G, the value $a_* = \bigsqcup_{n \geq 0} G^n(\bot)$ is the least fixpoint of G.

When the function is not ω-continuous, then we obtain a_* as above (we jump over a discontinuity) and then continue iterating. We then take the limit of such sequence, and the limit of limits etc., "ultimately" we obtain the fixpoint.

Patrick Cousot and Radhia Cousot (1979). Constructive versions of Tarski's fixed point theorems. *Pacific Journal of Mathematics*. 82:1: 43–57.

Another Lemma

Lemma: For an ω-continuous monotonic function G, let $a_* = \bigsqcup_{n \geq 0} G^n(\bot)$ and suppose that s' is such that $G(s') \sqsubseteq s'$. Then $a_* \sqsubseteq s'$.

Proof hint: $G^n(\bot) \sqsubseteq s'$.

Reachable Sets of States

Given a transition system $M = (S, \mathit{Init}, r, A)$, with $\bar{r} = \{(s, s') \mid (s, a, s') \in r\}$, consider the function $F : 2^S \to 2^S$ defined by

$$F(X) = \mathit{Init} \cup \bar{r}[X]$$

Then F is monotonic and ω-continuous (like a G function). Its least fixed point is the set of reachable states of M.

Reminder: $\bar{r}[X] = \{b \mid \exists a \in X.\ (a, b) \in \bar{r}\}$ (image of the set X under the relation $\bar{r}$)

Proving through Fixpoints of Approximate Functions

Meaning of a program (e.g. a set of reachable states) is a least fixpoint of F: $lfp(F)$.
Given specification s (e.g. $(pc = c) \rightarrow i \neq 42$), goal is to prove $\mathbf{lfp(F)} \subseteq \mathbf{s}$ (s is invariant)

- if $F(s) \subseteq s$ then $lfp(F) \subseteq s$ and we are done
- $lfp(F) = \bigcup_{k \geq 0} F^k(\emptyset)$, but that is too hard to compute because it is infinite union unless, by some luck, $F^{n+1}(\emptyset) = F^n$ for some n

Instead, we search for an inductive strengthening of s: find s' such that:

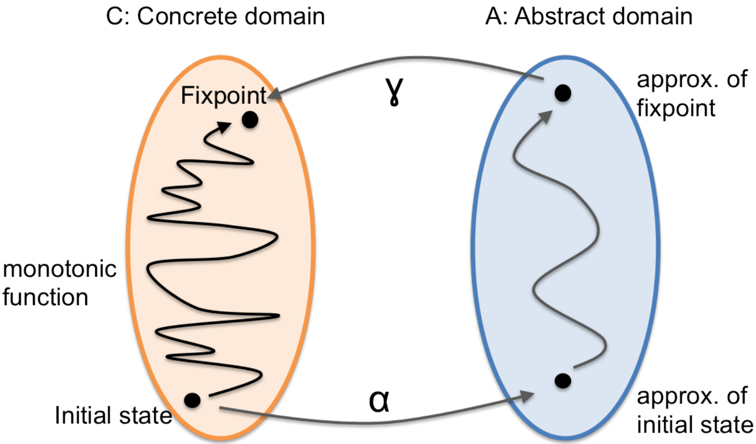
- $F(s') \subseteq s'$ (s' is inductive). If so, theorem says $lfp(F) \subseteq s'$
- $s' \subseteq s$ (s' implies the desired specification). Then $lfp(F) \subseteq s' \subseteq s$

How to find s' such that $F(s') \subseteq s'$? Iterating F is hard, so we try some simpler function $F_\#$:

- suppose $F_\#$ is *approximation*: $F(r) \subseteq F_\#(r)$ for all r
- we can find s' such that: $F_\#(s') \subseteq s'$ (e.g. $s' = F_\#^{n+1}(\emptyset) = F_\#^n(\emptyset)$)

Then: $F(s') \subseteq F_\#(s') \subseteq s'$. So, what we found is such that $s' \subseteq s$, we have $lfp(F) \subseteq s'$.
Abstract interpretation: automatically construct $F_\#$ using F and s

Abstract Interpretation Big Picture



Galois Connection

Galois connection (named after Évariste Galois) is defined by two monotonic functions $\alpha : C \to A$ and $\gamma : A \to C$ between partial orders $\leq$ on C and $\sqsubseteq$ on A, such that

$$\forall c, a. \quad \alpha(c) \sqsubseteq a \iff c \leq \gamma(a) \qquad (*)$$

(intuitively the condition means that c is approximated by a).

Lemma: The condition $(*)$ holds iff the conjunction of these two conditions:

$$\begin{array}{rcl} c & \leq & \gamma(\alpha(c)) \\ \alpha(\gamma(a)) & \sqsubseteq & a \end{array}$$

holds for all c and a.

Termination and Efficiency of Abstract Interpretation Analysis

Definition: A **chain** of length n is a sequence $s_0, s_1, \ldots, s_n$ such that

$$s_0 \sqsubset s_1 \sqsubset s_2 \sqsubset \ldots \sqsubset s_n$$

where $x \sqsubset y$ means, as usual, $x \sqsubseteq y \wedge x \neq y$

Definition: A partial order has a **finite height** n if it has a chain of length n and every chain is of length at most n.

A finite lattice is of finite height.

Example

The constant propagation lattice $\mathbb{Z} \cup \{\bot, \top\}$ is an infinite lattice of height 2. One example chain of length 2 is

$$\bot \sqsubset 42 \sqsubset \top$$

Here the γ function is given by

- $\gamma(k) = \ldots$ when $k \in \mathbb{Z}$
- $\gamma(\top) = \ldots$
- $\gamma(\bot) = \ldots$

The ordering is given by $a_1 \subseteq a_2$ iff $\gamma(a_1) \subseteq \gamma(a_2)$

Example

If a state of a (one-variable) program is given by an integer, then a concrete lattice element is a set of integers. This lattice has infinite height. There is a chain

$$\{0\} \subset \{0,1\} \subset \{0,1,2\} \subset \ldots \subset \{0,1,2,\ldots,n\}$$

for every n.

Convergence in Lattices of Finite Heignth

Consider a finite-height lattice $(L, \sqsubseteq)$ of height n and function

$$F : L \to L$$

What is the maximum length of sequence $\perp, F(\perp), F^2(\perp), \ldots$?
Give an effectively computable expression for $lfp(F)$.

Computing the Height when Combining Lattices

Let $H(L, \leq)$ denote the height of the lattice $(L, \leq)$.

Product

Given lattices $(L_1, \sqsubseteq_1)$ and $(L_2, \sqsubseteq_2)$, consider product lattice with set $L_1 \times L_2$ and potwise order

$$(x_1, x_2) \sqsubseteq (x'_1, x'_2)$$

iff ...

What is the height of the product lattice?

Exponent

Given lattice $(L, \sqsubseteq)$ and set V, consider the lattice $(L^V, \sqsubseteq')$ defined by

$$g \sqsubset' h$$

iff $\forall v \in V.g(v) \sqsubseteq h(v)$.

What is the height of the exponent lattice?

Computing the Height when Combining Lattices

Let $H(L, \leq)$ denote the height of the lattice $(L, \leq)$.

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Given lattices $(L_1, \sqsubseteq_1)$ and $(L_2, \sqsubseteq_2)$, consider product lattice with set $L_1 \times L_2$ and potwise order

$$(x_1, x_2) \sqsubseteq (x_1', x_2')$$

iff ...

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iff $\forall v \in V . g(v) \sqsubseteq h(v)$.

What is the height of the exponent lattice?

Answer: height of L times the cardinality of V.